A Width-One Corridor for Generalized Davenport Constants of C_5^3

Sze Chun Yiu
sze-chun.yiu@fysik.su.se

25 August 2026

Abstract

Let $D_k(G)$ be the least length that forces k pairwise disjoint nonempty zero-sum subsequences. We show that the generalized Davenport constants of C_5^3 satisfy $5k+10 \le D_k(C_5^3) \le 5k+11$ for every $k \ge 4$.

If $D_4(C_5^3) = 30$, then the lower line is exact for every $k \ge 2$. To analyze the remaining alternative, we study a hypothetical total-zero sequence of length 31 with no nonempty zero-sum subsequence of length at most five. For saturated short-free sequences over elementary abelian groups of odd prime exponent, we prove a defect certificate that excludes multiplicity $p-2$. In the case $p=5$, only multiplicities $1,2,4$ remain. If s is support size and c_i counts multiplicity i, then $c_2 = 31 - s - 3c_4$ and $c_1 = 2s - 31 + 2c_4$.

The subsequence of points of multiplicity at least two has length $62 - 2s - 2c_4$. Using $\eta(C_5^2) = 13$, we prove that this stratum must span rank three whenever $s + c_4 \le 24$. We also prove an exact identity coupling support deficit, repetitions inside zero-sum atoms, and overlaps between atom supports. Together with a theorem-grade exclusion through support ten, these results isolate the residual obstruction phase without deciding it.

The exact value remains $D_4(C_5^3) \in \{30, 31\}$. A larger bounded search through support 22 is reported only as bounded computational evidence and is not used as theorem authority.

Keywords: generalized Davenport constants; zero-sum sequences; elementary abelian groups; short zero sums; inverse zero-sum problems

1. Introduction

Generalized Davenport constants measure the sequence length required to force several pairwise disjoint zero sums. Rank-two groups exhibit strong eventual structure, whereas rank-three groups can depart from the expected arithmetic progression.

For C_5^3, the exact values $D_2 = 20$ and $D_3 = 25$, together with known short-zero-sum thresholds, reduce the next value to

$$D_4(C_5^3) \in \{30, 31\}.$$

This one-bit ambiguity controls the tail: the lower candidate $D_4 = 30$ forces $D_k = 5k + 10$ for every $k \ge 2$. The upper candidate would yield a rigid length-31 total-zero sequence with no zero sum of length at most five.

We combine the recurrence with symbolic analysis of that hypothetical obstruction. The purpose is twofold: to obtain a rigorous width-one theorem for the full tail and to identify exactly where existing rank reductions cease to be automatic. Computation is kept in a separate evidence class.

Our contributions are:

1. the width-one corridor for every $k \geq 4$;
2. the conditional exact tail if $D_4 = 30$;
3. an odd-prime saturation-defect lemma excluding multiplicity $p - 2$;
4. a theorem-grade support-at-least-eleven result for the length-31 obstruction;
5. an exact multiplicity parameterization for every remaining support stratum;
6. the rank-forcing phase $s + c_4 \leq 24$ for the high-multiplicity stratum; and
7. an identity coupling support deficit to internal atom repetition and cross-atom overlap.

2. Definitions and known inputs

For a finite abelian group G, $D_k(G)$ is the least integer n such that every sequence of length at least n over G contains k pairwise disjoint nonempty zero-sum subsequences. Let $s_{\leq \ell}(G)$ denote the least length forcing a nonempty zero-sum subsequence of length at most ℓ.

We use the generalized-Davenport recurrence framework [1]

$$D_{k+1}(G) \leq \max\{D_k(G) + \ell,\ s_{\leq \ell}(G) - 1\},$$

and the known lower bound $D_k(C_5^3) \geq 5k + 10$ in the range considered here. The exact values used below are

$$D_2 = 20, \qquad D_3 = 25, \qquad s_{\leq 6} = 24,$$

while $s_{\leq 5}(C_5^3) = \eta(C_5^3) = 33$. We also use the rank-two value $\eta(C_5^2) = 13$.

3. A width-one tail

Taking $\ell = 6$ gives

$$D_4 \leq \max\{25 + 6, 24 - 1\} = 31.$$

Together with the lower bound, this proves $D_4 \in \{30, 31\}$.

Theorem 1. For every $k \geq 4$,

$$5k + 10 \leq D_k(C_5^3) \leq 5k + 11.$$

Proof. The base case is $D_4 \leq 31$. Apply the recurrence with $\ell = 5$ and $s_{\leq 5} - 1 = 32$. If $D_k \leq 5k + 11$, then both terms in the maximum are at most $5(k + 1) + 11$. The known lower bound supplies the lower line. □

Theorem 2. If $D_4(C_5^3) = 30$, then

$$D_k(C_5^3) = 5k + 10 \qquad (k \geq 2).$$

Proof. The statement holds for $k = 2, 3, 4$. Applying the same recurrence with $\ell = 5$ propagates the upper bound $5(k+1) + 10$, which matches the known lower bound. □

There is no converse here: the assumption $D_4 = 31$ does not by itself prove that later terms lie on the upper line.

4. The hypothetical length-31 obstruction

Assume $D_4(C_5^3) = 31$. The extremal characterization of Freeze and Schmid provides a total-zero sequence S of length 31 whose maximum factorization length is four. If M is the minimum length of an atom dividing S, their recurrence gives $31 = D_4 \leq D_3 + M = 25 + M$, so $M \geq 6$. On the other hand, $s_{\leq 6}(C_5^3) = 24$ forces every 31-term sequence to contain a zero-sum subsequence of length at most six; an inclusion-minimal such subsequence is an atom. Hence $M = 6$, and S contains no zero-sum subsequence of length at most five. Thus the upper candidate yields a sequence satisfying

$$|S| = 31, \qquad \sigma(S) = 0,$$

with no nonempty zero-sum subsequence of length at most five. Every element has multiplicity at most four. If its support exceeds eight, it is saturated: otherwise a short-free one-term extension would have length 32, whereas Property C for C_5^3 forces every length-32 short-free sequence to have exactly eight support points.

Theorem 3 (saturation defect). Let p be odd and let S be a saturated p-short-free sequence over an elementary abelian exponent-p group. If a nonzero element x has multiplicity $m < p$, then there is a subsequence R such that

$$|R| \leq p - 1 - m, \qquad x \notin \mathrm{supp}(R), \qquad \sigma(R) = -(m+1)x.$$

Proof. Append one further copy of x. Saturation supplies a zero-sum subsequence of length at most p that uses the appended occurrence. It must also use every original copy of x; otherwise replacing the appended copy by an unused original occurrence would give a short zero sum already contained in S. Removing the $m + 1$ copies of x leaves the required R. □

Corollary 4. Multiplicity $p - 2$ is impossible.

Indeed, the defect subsequence would have length at most one and sum x, forcing a forbidden additional copy of x. For $p = 5$, multiplicity three is absent. Singleton and double points also acquire defect certificates of length at most three and two, respectively.

5. Exact multiplicity grammar

Let $s = |\,\mathrm{supp}(S)|$, and let c_1, c_2, c_4 count points of multiplicity one, two, and four. Then

$$c_1 + c_2 + c_4 = s, \qquad c_1 + 2c_2 + 4c_4 = 31.$$

Writing $c_4 = j$ gives

$$c_2 = 31 - s - 3j, \qquad c_1 = 2s - 31 + 2j.$$

Thus c_1 is odd, and nonnegativity determines the complete admissible range for every support size. For example, the support-23 patterns are

$$1^{15}2^8, \qquad 1^{17}2^5 4, \qquad 1^{19}2^2 4^2,$$

and the support-24 patterns are

$$1^{17}2^7, \qquad 1^{19}2^4 4, \qquad 1^{21}24^2.$$

6. The rank-forcing phase

Let H be the subsequence consisting of all points of multiplicity two or four. Its length is

$$|H| = 2c_2 + 4c_4 = 62 - 2s - 2c_4.$$

If H were contained in a subgroup of rank at most two, it would be a 5-short-free sequence over C_5^2. The value $\eta(C_5^2) = 13$ forces a short zero sum once $|H| \geq 13$. Since $|H|$ is even, the contradiction begins at $|H| \geq 14$.

Theorem 5 (rank-forcing phase). The high-multiplicity stratum spans rank three whenever

$$s + c_4 \leq 24.$$

Proof. The condition $|H| \geq 14$ is equivalent to $62 - 2s - 2c_4 \geq 14$, hence to $s + c_4 \leq 24$. □

The theorem gives the following exact boundary:

Support	c_4	Consequence from Theorem 5
23	0 or 1	rank three forced
23	2	threshold is silent
24	0	rank three forced
24	1 or 2	threshold is silent
at least 25	any admissible value	threshold is silent

Silence is not a converse. Outside the phase, rank three may follow from a different argument.

7. Theorem-grade low-support exclusion and bounded evidence

Symbolic reductions combined with two exact state representations exclude every support size through ten.

Theorem 6. Every length-31 total-zero 5-short-free sequence over C_5^3, if one exists, has support at least eleven.

Computer-assisted proof. Multiplicity at most four first excludes supports at most seven. At support eight, the only length pattern is $4^7 3$. Adding the missing fourth copy of the triple point preserves short-freeness; total sum then forces that point to equal the negative support sum. After normalizing a basis by $\mathrm{GL}(3, 5)$, two exact subset-sum representations enumerate all 564 normalized supports and find none satisfying this condition.

For supports nine and ten, saturation and Corollary 4 exclude multiplicity three. The support-nine equations leave only $4^7 21$; a canonical search with the last point forced by total sum visits 6,537,270 states in each of two independent exact-weight subset-sum engines and finds no solution. At support ten, the only patterns are $1^3 4^7$ and $1, 2^3 4^6$. Four multiplicity-four points must span rank three because 16 terms in a rank-two subgroup would contradict $\eta(C_5^2) = 13$. Normalizing an independent triple to the standard basis makes the remaining enumeration complete. The two engines agree exactly: they visit respectively 210,700 states with 3,558 leaves and 272,119 states with no leaves, finding zero solutions in both patterns.

Both engines reject a branch exactly when adding a term creates a zero-sum subsequence of one of the lengths one through five; one stores explicit byte reachability by weight and group sum, while the other stores translation masks in a 128-bit representation. Their source, build instructions, expected rows, and machine-readable results are included as ancillary files. Thus every support stratum through ten is exhausted. □

A larger bounded computation found no such sequence through support 22. That search has not received an external independent replay and is not used in any proof above or below. Its bounded conclusion is useful for prioritizing the next case split, but it does not establish support at least 23 as a theorem.

8. Atom repetition and overlap

Under the conditional assumption $D_4 = 31$, extremal factorization arguments force one of the atom-length types

$$(6, 6, 6, 13) \qquad \text{or} \qquad (6, 6, 7, 12).$$

Let $S = U_1 \cdots U_r$ be an occurrence-disjoint atom factorization. For each group element g, let r_g be the number of atom supports containing g, and define the internal deficit

$$\delta_i = |U_i| - |\operatorname{supp}(U_i)|.$$

Theorem 7 (repetition-overlap identity). Every occurrence-disjoint factorization satisfies

$$|S| - |\operatorname{supp}(S)| = \sum_i \delta_i + \sum_{g \in \operatorname{supp}(S)} (r_g - 1).$$

Proof. Summing the atom support sizes counts every global support point r_g times. Hence

$$\sum_i \delta_i = |S| - \sum_g r_g,$$

while cross-atom overlap equals

$$\sum_g r_g - |\operatorname{supp}(S)|.$$

Adding the two identities proves the claim. □

Thus any independently established lower support bound s_0 gives a total internal-repetition plus cross-overlap budget of at most $31 - s_0$. The proven value $s_0 = 11$ gives budget 20. If the bounded support-22 computation is later independently certified, the budget sharpens to eight; that sharper value is conditional and not used as theorem authority here.

9. The residual obstruction phase

A decisive next analysis should impose simultaneously:

1. the multiplicity pattern from Section 5;
2. the rank status from Theorem 5;
3. the saturation-defect certificate at every singleton and double point;
4. one of the two conditional atom-length types;
5. the repetition-overlap identity; and
6. total sum zero with exclusion of zero sums of lengths one through five.

This intersection is substantially smaller than an unstructured support search. A surviving sequence would require independent factorization verification before establishing $D_4 = 31$. A complete, independently replayed infeasibility proof for the length-31 target would establish the lower candidate $D_4 = 30$.

10. Relation to prior work

The generalized-Davenport recurrence, lower-bound framework, extremal factorization language, and eventual-linearity context are established results [1]. The C_0 framework and short-zero-sum localization are also prior work [2], as are the inverse zero-sum program [3] and rank-two inverse results [4].

The residual contribution is the width-one synthesis for C_5^3, the saturation-defect specialization, the exact multiplicity grammar, the rank-forcing phase, and the coupling of support deficit with atom repetition and overlap. These are structural compression results; they do not decide the final extremal bit.

11. Reproducibility and limitations

The corridor, defect lemma, and support-ten exclusion are separately bound to exact records. An independent verifier enumerates all multiplicity patterns for supports 8-31, checks the rank-phase equivalence, reproduces the support-23 and support-24 tables, and validates the atom identity on independently generated factorizations. The displayed proofs carry all-parameter authority.

The exact value of $D_4(C_5^3)$ remains open, as does the associated length-31 extremal-spectrum statement. The support-22 frontier is bounded computation pending external replay. The rank phase is one-way, and the atom identity is a compression principle rather than a contradiction. No additional implementation can substitute for an independently auditable final obstruction proof.

12. Conclusion

The generalized Davenport constants of C_5^3 remain within one unit of the line $5k+10$ for every $k \geq 4$, and the lower choice at $k = 4$ would fix the entire tail. The hypothetical upper-line obstruction has substantially more structure than this corridor alone reveals: its multiplicities have an exact grammar, its high-multiplicity stratum enters a sharp rank-forcing phase, and its atom factorization obeys a global repetition-overlap budget.

These results explain where the residual difficulty begins and define a smaller intersection problem for the unresolved extremal bit. They do not claim to have resolved it.

Data and code availability

The verification package contains exact multiplicity enumeration, rank-phase checks, atom-identity tests, and separately labeled bounded-search records. A permanent archival identifier should be added before final submission.

References

1. M. Freeze and W. A. Schmid, "Remarks on a Generalization of the Davenport Constant," *Discrete Mathematics* **310**, 3373-3389 (2010). DOI: 10.1016/j.disc.2010.07.028
2. Y. Fan, W. Gao, G. Wang, Q. Zhong, and J. Zhuang, "On Short Zero-Sum Subsequences of Zero-Sum Sequences," *Electronic Journal of Combinatorics* **19**(3), P31 (2012). DOI: 10.37236/2602
3. W. Gao, A. Geroldinger, and W. A. Schmid, "Inverse Zero-Sum Problems," *Acta Arithmetica* **128**, 245-279 (2007). DOI: 10.4064/aa128-3-5
4. Q. Zhong, "On the Inverse Problem of the k-th Davenport Constants for Groups of Rank 2," *Combinatorica* **45**, article 31 (2025). DOI: 10.1007/s00493-025-00153-3